

## Derivatives

**Goals:** Define the Derivative and apply it to initial examples

**Before Class Videos:** 2.1.1, 2.1.2

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**Example 1:** Recall the velocity problem from the first day of class. If Trefor has some distance function  $f(t)$ , how fast is he going at 10 minutes? Let's explore graphically and numerically:

**Definition:**

The derivative of  $f(x)$  at  $c$  is given by

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If it does, we say  $f$  is differentiable at  $c$ .

**Example 2:** Compute the derivative of  $f(x) = x^2$  at  $x = 2$ .

**Definition:**

The derivative of  $f(x)$  on an open interval  $I$  is the function  $f'(x)$  given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists for all  $x$  in  $I$ . If it does we say  $f(x)$  is differentiable on  $I$ .

**Example 3:**  $f(x) = 1/x^2$ .

**Example 4:**  $f(x) = \sqrt{x}$ .

**Example 5:** Let's explore if the following functions are differentiable at  $x = 0$ :

a)

$$f(x) = \begin{cases} 2, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

b)  $f(x) = |x|$

c)  $f(x) = x^{2/3}$

d)  $f(x) = x^{1/3}$